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MIL-STD-3069(NAVY)
11 October 2022

DEPARTMENT OF DEFENSE
DESIGN CRITERIA STANDARD

SPECIALIZED REUSABLE METAL CONTAINER DESIGN FOR NAVAL AVIATION REPAIRABLES (NON-ORDNANCE)

AMSC 10360

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FOREWORD

1. This standard is approved for use by the Department of the Navy.

2. This standard was developed using MIL-STD-648E as the basis for the tailored requirements specified herein. The requirements are for specialized reusable metal container design development for non-ordnance naval aviation repairables only.

3. For technical guidance on the contents of this document or to submit technical changes or recommended changes, email karen.a.mcdonnell1.civ@us.navy.mil, Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, (Code BL32600), Bldg. 333, Route 547, Joint Base MDL, NJ 08733 or email matthew.l.fee.civ@us.navy.mil, Commanding Officer, NAVSUP Weapon Systems Support, Attn: Code N241, 700 Robbins Avenue, Philadelphia, PA 19111-5098.

4. Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center, Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to LKE-NAVAIRDocCom.fct@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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CONTENTS

Paragraph Page

1. SCOPE 1

1.1 SCOPE 1
1.2 PURPOSE 1
1.3 APPLICABILITY 1

2. APPLICABLE DOCUMENTS 1

2.1 GENERAL 1
2.2 GOVERNMENT DOCUMENTS 1
2.2.1 Specifications, standards, and handbooks 1
2.2.2 Other Government documents, drawings, and publications 2
2.3 NON-GOVERNMENT PUBLICATIONS 3
2.4 ORDER OF PRECEDENCE 4

3. DEFINITIONS 5

3.1 GENERAL 5
3.1.1 AMBIENT (TEMPERATURE) 5

4. GENERAL REQUIREMENTS 5

4.1 GENERAL CONTAINER REQUIREMENTS 5
4.1.1 Weight and cube 5
4.1.2 Temperature stability 5
4.1.3 Pressure and vacuum retention 5
4.1.4 Drainage 6
4.1.5 Standard parts 6
4.1.6 Container structure material 7
4.1.6.1 Aluminum 7
4.1.6.1.1 Aluminum cleaning 7
4.1.6.2 Stainless steel 7
4.2 FIT AND COMPATIBILITY 7
4.3 DYNAMIC CHARACTERISTICS 7

5. DETAIL REQUIREMENTS 8

5.1 COVER 8
5.1.1 Stacking feature 8
5.1.2 Sealing 8
5.1.3 Service and maintenance facilities 8
5.1.3.1 Breather valve 8
5.1.3.1.1 Cracking and sealing pressure 9
5.1.3.1.2 Flow rate 9
5.1.3.2 Desiccant basket 9
5.1.3.3 Relative humidity indicator 9

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

5.1.3.4 Visual inspection ports *9*
5.1.3.5 Records receptacle *9*
5.1.3.6 Leak test valve *10*
5.2 BASE 10
5.2.1 Mounting frame *10*
5.2.1.1 Restraint conditions *10*
5.2.1.2 Mounting frame - VERTREP *10*
5.2.2 Suspension system *10*
5.2.2.1 Shock isolator markings *10*
5.2.2.2 Cushioning materials *10*
5.2.3 Fasteners *11*
5.2.4 Storage box (if required) *11*
5.2.5 Orientation/security seal *11*
5.2.6 Drain plug *11*
5.2.7 Skids *11*
5.2.8 Roller interface *12*
5.3 IDENTIFICATION 12
5.3.1 Markings *12*
5.3.2 Shipping information *13*
5.3.3 Nameplate *13*
5.3.4 Container serial number *14*
5.4 INSTALLATION INSTRUCTIONS 14
5.5 TRANSPORTABILITY 15
5.5.1 Transfer-at-sea *15*
5.5.1.1 CONREP *15*
5.5.1.2 VERTREP *15*
5.6 PROVISIONS 15
5.6.1 Handle provision characteristics *15*
5.6.2 Lifting provision characteristics *16*
5.6.3 Tiedown provisions *17*
5.6.4 Towing provisions *17*
5.6.5 Pass/fail criteria *17*
5.7 SHORE BASED 17
5.7.1 Forklift capability *18*
5.7.2 MK 45 handlift truck interface *18*
5.7.3 Pallet jack interface *18*
5.8 MASS SIMULATOR DESIGN 18
5.9 FINISH 18
5.9.1 Aluminum *19*
5.9.2 Surface finishing *19*
5.9.3 Plating *19*
5.9.4 Stainless steel *19*
5.10 WORKMANSHIP 19
5.10.1 Welding *19*
5.10.1.1 Visual inspection criteria for aluminum welds *19*

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

5.10.1.2 Visual inspection criteria for steel welds *20*
5.10.1.3 Cleaning *20*
5.10.1.4 Deoxidizing *21*
5.10.1.5 Chemical treatment *21*
5.10.1.6 Mechanical means *21*
5.11 VERIFICATION 21

6. NOTES 26

6.1 INTENDED USE 26
6.2 ACQUISITION REQUIREMENTS 26
6.3 ASSOCIATED DATA ITEM DESCRIPTIONS (DIDS) 26
6.4 INTERNATIONAL STANDARDIZATION AGREEMENT IMPLEMENTATION 27
6.5 SUBJECT TERM (KEY WORD) LISTING 27

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

APPENDIX CONTENTS

Paragraph Page

A.1 SCOPE 29
A.1.1 SCOPE 29
A.2 APPLICABLE DOCUMENTS 29
A.2.1 GENERAL 29
A.2.2 SPECIFICATIONS AND STANDARDS 29
A.2.3 NON-GOVERNMENT PUBLICATIONS 29
A.3 VERIFICATION AND TEST REQUIREMENTS 30
A.3.1 MASS SIMULATOR 30
A.3.1.1 Instrumentation 30
A.3.2 INSPECTION 30
A.3.3 LEAKAGE INTEGRITY 30
A.3.3.1 Leak test 30
A.3.3.2 Vacuum test 31
A.3.4 FIT AND COMPATIBILITY TEST 31
A.3.5 VIBRATION 31
A.3.5.1 Sine vibration 32
A.3.5.1.2 Sine dwell 32
A.3.5.2 Random vibration 32
A.3.5.2.1 Truck transportation 32
A.3.5.2.2 Underslung cargo 32
A.3.6 MECHANICAL SHOCK 32
A.3.6.1 Cornerwise-drop (rotational) test 32
A.3.6.2 Edgewise-drop (rotational) test 33
A.3.6.3 Impact test 33
A.3.6.4 Transfer-at-sea shock test 33
A.3.6.5 Flat bottom drop 33
A.3.6.6 Repetitive shock test 33
A.3.7 TIPOVER TEST 33
A.3.8 LOAD STABILITY 33
A.3.9 SUPERIMPOSED LOAD 34
A.3.10 LIFTING FITTINGS, CLOSURE FITTINGS, AND TIEDOWN ATTACHMENT POINTS 34
A.3.10.1 Lifting fittings interface test 34
A.3.10.2 Lifting fittings strength test 34
A.3.10.4 Tiedown strength test 34
A.3.10.5 Closure fitting strength test 34
A.3.11 MOUNT AND STRUCTURE LOAD TEST 35
A.3.12 FORKLIFT TRUCK COMPATIBILITY TEST 35
A.3.13 HANDLIFT TRUCK MK 45 COMPATIBILITY 35
A.3.13.1 Static overload 35

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T02:32Z
Check the source to verify that this is the current version before use.

A.3.13.2 Shock test *35*
A.3.13.3 Rolling test *35*
A.3.14 PALLET JACK COMPATIBILITY TEST 36

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Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard establishes requirements for designing and testing specialized reusable metal shipping and storage container(s) intended to protect high-dollar naval aviation repairable assemblies in the naval distribution system. This standard is only applicable for non-ordnance related items in support of Naval Aviation. The container must provide watervaporproof, shock, vibration, and physical protection for the item during shipping, handling and storage.

1.2 Purpose. The purpose of this standard is to provide criteria in designing a specialized shipping and storage container. The container must protect the item from moisture, shock, vibration and physical damage during domestic and overseas shipment and during storage in the naval environment. The container must be designed to be compatible with Connected Replenishment (CONREP), Vertical Replenishment (VERTREP), over-the-road and air transportability. Design criteria and testing must be in accordance with requirements noted below. The container must have a minimum operational life of twenty (20) years with normal maintenance.

1.3 Applicability. This standard is applicable to the Department of the Navy and its contractors.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this standard. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

STANAG 2828	-	Military Pallets, Packages and Containers
STANAG 2829	-	Materials Handling Equipment

COMMERCIAL ITEM DESCRIPTIONS

A-A-208	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-56032	-	Ink, Marking, Epoxy Base
A-A-59136	-	Cushioning Material, Packaging, Closed Cell Foam Plank

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	-	Chemical conversion Coatings on Aluminum and Aluminum Alloys
MIL-DTL-15024		Plates and tags for Identification of Equipment, General Specification for
MIL-PRF-6855	-	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for
MIL-PRF-9954	-	Glass Beads: for Cleaning and Peening
MIL-W-22248	-	Weldments, Aluminum and Aluminum Alloy
MIL-DTL-27166	-	Valve, Pressure Equalizing, Gaseous Products
MIL-PRF-81733	-	Sealing and Coating Compound, Corrosion Inhibitive

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-209	-	Lifting and Tiedown Provisions
MIL-STD-648	-	Specialized Shipping Containers
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Galvanic Compatibility of Electrically Conductive Materials
MIL-STD-913	-	Requirements for the Certification of Sling Loaded Military Equipment for External Transportation by Department of Defense Helicopters
MIL-STD-1366	-	Transportability Criteria
MIL-STD-1472	-	Human Engineering
MIL-STD-1791		Interface Standard
MIL-STD-2073-1	-	Standard Practice for Military Packaging
MIL-STD-3010	-	Test Procedures for Packaging Materials and Containers
MIL-STD-3028	-	Joint Modular Intermodal Container
MIL-STD-31000	-	Technical Data Packages

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATION

49 CFR Parts 100-180	-	Transportation

(Copies of this document are available at www.ecfr.gov.)

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NAVAL AIR SYSTEMS COMMAND

TECHNICAL MANUAL

NAVAIR A1-V22AB-CLG-000 - Cargo Handling Manual V-22 Tiltrotor

(Copies of this document are available online at https://airworthiness.navair.navy.mil/ and https://mynatec.navair.navy.mil/ accessible via account.)

NAVAL SEA SYSTEMS COMMAND

NAVSEA DRAWINGS

2642780 - Container Capability with Handlift Trucks MK42/MK 45

(Copies of this document are available https://webjedmics.navair.navy.mil.)

NAVY TACTICS, TECHNOLOGIES AND PROCEDURES (NTTP)

NWP 4-01.4 - Underway Replenishment

(Copies of this document are available online at https://maritime.org/doc/pdf/unrep-nwp04-01.pdf.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A153/A153M	-	Standard Specification for Zinc Coating (Hot Dip) on Iron and Steel
ASTM A380/A380M	-	Standard Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems
ASTM A514/A514M	-	Standard Specification for High-Yield Strength, Quenched and Tempered Alloy Steel Plate, Suitable for Welding
ASTM A517/A517M	-	Standard Specification for Pressure Vessel Plates, Alloy Steel, High Strength, Quenched and Tempered
ASTM A967/A967M	-	Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM B841	-	Standard Specification for Electrodeposited Coatings of Zinc Nickel Alloy Deposits
ASTM D991	-	Standard Test Method Rubber Property-Volume Resistivity of Electrically Conductive and Antistatic Products

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ASTM E165/E165M	-	Standard Practice for Liquid Penetrant Testing for General Industry
ASTM E1444/E1444M	-	Standard Practice for Magnetic Particle Testing Systems, Equipment and Facilities

(Copies of these documents are available online at http://www.astm.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	-	Structural Welding Code – Steel
AWS D1.2/D1.2M	-	Structural Welding Code – Aluminum
AWS D1.3/D1.3M	-	Structural Welding Code – Sheet Steel
AWS D1.6/D1.6M	-	Structural Welding Code – Stainless Steel

(Copies of these documents are available online at https://aws.org.)

ELECTROSTATIC DISCHARGE ASSOCIATION (ESDA)

ANSI/ESD STM11.11	-	Standard Test Method for Protection of Electrostatic Discharge Susceptible Items - Surface Resistance Measurement of Planar Materials

(Copies of this document are available online at https://www.esda.org.)

SAE INTERNATIONAL

SAE AMS2700	-	Passivation of Corrosion Resistant Steels
SAE AMS3269	-	Sealing Compound, Polysulfide (T) Synthetic Rubber for Integral Fuel Tank and Fuel Cell Cavities Low Density (1.20 to 1.35 sp gr), for Intermittent Use to 360 °F (182 °C)
SAE AMS-STD-595	-	Colors Used in Government Procurement
SAE AS5017	-	Air Filling Valves
SAE AS5135	-	Desiccant Port and Desiccant Holder
SAE AS5389	-	Pallet Lift Truck and Container Fork Pocket Interface
SAE AS26860	-	Indicator, Humidity, Plug, Color Change
SAE AS27166	-	Valve, Pressure Equalizing, Gaseous Products

(Copies of these documents are available online at http://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 General. The terms used throughout this standard are as follows. Terms related to container design are in accordance with MIL-STD-648. Terms related to packaging are in accordance with MIL-STD-2073-1. Hazardous materials (HAZMAT) terms are defined in 49 CFR Parts 100-180.

3.1.1 Ambient (temperature). Defined as 77°±18 °F for the purposes of this standard.

4. GENERAL REQUIREMENTS

4.1 General container requirements. The container shall be designed in accordance with the requirements of this section, Section 5, Appendix A and with the following criteria:

a. Minimize weight and cube (see 4.1.1).
b. Protect the item from environmentally induced corrosion, deterioration physical and mechanical damage during storage, handling and shipment.
c. Emphasize the use of corrosion-resistant aluminum as the primary material for optimum performance, reduced maintenance and improved longevity characteristics.
d. Use steel, plastic and other non-metal materials for internal support frames only as necessary to minimize cost and meet performance requirements.
e. When placed in a stacked configuration no permanent deformation of the container cover in excess of 0.75 inch, or any deformation that inhibits the removal of the cover, item or contacts the item.
f. Meet Department of Transportation requirements for truck, rail, air, and ship loading.
g. Minimize transportation costs without sacrifice to safety and container integrity per MIL-STD-1366.
h. Comply with transportability dimensional constraints and weight requirements per MIL-STD-1366.
i. Have a minimum operational life of twenty (20) years with normal maintenance.

The developed specialized reusable container design shall be documented as specified on the Contract Data Requirements List (CDRL) and a Product Level Technical Data Package (TDP) shall be delivered (see 6.3).

4.1.1 Weight and cube. The container shall be designed for minimum weight and cube consistent with protection of the item, durability, intended use, economy, safety, transportability, and interfacing equipment.

4.1.2 Temperature stability. The container assembly shall be designed to perform satisfactorily between 160 °F (71 °C) and -20 °F (-29 °C).

4.1.3 Pressure and vacuum retention. The container shall be able to withstand a differential pressure setting equal to 1.0 psig (±0.05) and -1.0 psig (±0.05).

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4.1.4 Drainage. Provisions shall be incorporated to ensure that every pocket on the exterior of the container is capable of draining in the normal storage position of the container. When practicable, holes shall be drilled of sufficient size to allow proper draining, without affecting the leak integrity of the container. At faying surfaces, discontinuous welds, or other areas where drilling is impractical, the openings shall be filled with a sealing compound in accordance with MIL-PRF-81733 or SAE AMS3269 in areas where water intrusion results in damage to the container. Areas where sealing compound is used shall be indicated on the drawing in accordance with MIL-STD-31000.

4.1.5 Standard parts. Standard parts shall be used unless they are technically or economically impractical. Any use of any other parts, or variations of parts, other than those listed below, shall be approved by the design and procuring activities. The listing of common standard parts for use in appropriate applications for Navy containers is listed in Table I.

TABLE I. Common standard parts for Navy containers.

Description	Non-government standard	Navy drawing
Access port		6212707
Records holder (receptacle)		799AS105 5166322
Latch		6212861 6213102 6213873
Desiccator	SAE AS5135	6212868
Humidity indicator plug	SAE AS26860	5166628
Breather valve	SAE AS27166	6212863
Schrader valve	SAE AS5017	-
Observation window		6212706
Handle		6212630
Lift rings		6214106
Shackle		6212862
Fork pocket 2/ and pallet jack	SAE AS5389	5167633 5167693
Endlift extrusion		5167399
Endlift-adapter		5167401
Center lift		3214AS110

NOTES:

1/ For a complete listing of standard parts, contact NSWC IHEODTD, Detachment Picatinny, Code G13 Whittemore Ave., Building 458, Floor 1, Picatinny, NJ, 07806.

2/ For designs required to interface with NATO interoperability, reference STANAG 2828 and STANAG 2829.

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4.1.6 Container structure material.

4.1.6.1 Aluminum. Unless otherwise specified, aluminum alloys 5052-H34/H38, 5454-H32, 6061-T6 or 6063-T52 shall be used. Use of any other alloys require written technical justification and approval from the procuring activity's Packaging, Handling, Storage & Transportation Branch or Department, as appropriate. Combinations of these alloys is acceptable depending on required qualities of extrudability, machinability, fabrication techniques and compatibility. However, dissimilar couples, as defined in MIL-STD-889, shall be avoided.

4.1.6.1.1 Aluminum cleaning. Aluminum shall be cleaned in accordance with MIL-DTL-5541, except that uninhibited alkaline solution shall not be used. Further chemical finish, chemically conversion coating or painting is not required.

4.1.6.2 Stainless steel. All threaded or working ferrous surfaces, if any, shall be corrosion resistant (stainless) steel, 300 series or 18-8. The selected series shall be determined by the degree of corrosion resistance required by the specific application, while the material condition shall be based on the mechanical properties required of the components. Care shall be taken to prevent galling.

4.2 Fit and compatibility. The container shall be designed to:

a. Ensure the item fits securely without interference or binding.
b. Accommodate item sensitivities and special shipping requirements, to include but not limited to, attitude, electromagnetic interference field forces, electrostatic discharge, humidity or temperature, as specified (see 6.2).
c. Facilitate simple process loading and unloading using typical material handling equipment.
d. Meet the fit and compatibility test and tipover test of A.3.4 and A.3.7, respectively. A lack of "fit" or extreme awkwardness during the necessary loading and unloading process is cause for rejection.
e. Meet loading and unloading times, if specified (see 6.2). Not meeting specified loading or unloading times with the specified number of appropriately qualified personnel is cause for rejection.
f. Failure of the container to remain upright when tipped is cause for rejection.
g. Meet shock and vibration tests of A.3.5 and A.3.6.
h. Meet handling tests of A.3.8 – A.3.14.

4.3 Dynamic characteristics. Materials for use in shock or vibration attenuation systems shall protect to the approved shipping/non-operational fragility limit of the item. Vibration testing of A.3.5 and shock testing of A.3.6 shall be satisfied (see 6.2). During and after testing, evidence of one or more of the following shall be cause for rejection:

a. Damage to the item (visual or functional).
b. Failure of the suspension system to protect the item to the approved shipping/non-operational fragility limit. The response of the complete measuring system from accelerometer through the readout instrument shall be filtered. Data shall be filtered at a

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minimum of 90% over the frequency range of 2 Hz (lower limit) to the upper limit equal to 15 (± 5%) times the resonant frequency of the shock isolation system.

c. Permanent deformation of any portion of the container.
d. Structural failure of the isolation system.
e. Failure of the isolation system to meet random vibration fragility level requirements.
f. Loosening of the restraining material or devices.
g. Evidence of the item striking the container walls.
h. Leakage in excess of the criteria in A.3.3.
i. Excessive rotation or shifting of the item that could cause damage or prohibit removal of the item by the intended means.
j. Failure of the container to retain the item.
k. Failure of the container to permit continued safe handling.

Note: Repairable weld cracks in other than principal load paths that do not result in failure, as stipulated above, are not cause for rejection.

5. DETAIL REQUIREMENTS

5.1 Cover.

5.1.1 Stacking feature. The container shall be designed to permit the stacking of like containers, or to support other loading without structural failure of the stacking features or damage to the item. Testing shall be in accordance with A.3.8 and A.3.9 and shall meet the following criteria:

a. Stacked containers shall not interfere with forklift entry.
b. Stacking features shall not interfere with a safe and level stack of like containers.
c. Stacking features shall limit lateral and longitudinal movement during a stack of like containers. Slippage shall not be in excess of the stacking provisions.

5.1.2 Sealing. The container closure shall be a gasket tongue-in-groove, with the groove portion located in the container cover. The closure shall incorporate a feature to protect the sealing surfaces of the tongue and the gasket. The loading force on the gasket shall be controlled to prevent compression in excess of 30 percent. The gasket conforming to MIL-PRF-6855, Class 2, Type B, Grade 40 shall be neoprene and easily replaceable. Separately applied sealing compounds or materials shall not be allowed.

5.1.3 Service and maintenance facilities. All service facilities shall be grouped at one end of the container. Consistent with these requirements, the relative humidity indicator shall be as far as practical from the desiccant receptacle. The breather valve shall be placed as close as is practical to the center of the end of the container where the facilities are grouped and shall be located to introduce air into the container by way of the desiccant receptacle. All service and maintenance facilities shall be recessed or protected by ribs on the container shell.

5.1.3.1 Breather valve. An automatic pressure and vacuum relief valve with recessed manual pressure release conforming to MIL-DTL-27166 shall be provided to protect the item

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from moisture and the container from excessive pressure differentials. See Table I for Navy standard breather valve.

5.1.3.1.1 Cracking and sealing pressure. The valve shall crack at a pressure differential of 1.0 ±0.05 psi and shall reseal at a minimum pressure differential of 0.5 psi for both vacuum and pressure relief. This pressure shall be used to perform the leak and vacuum tests of A.3.3.1 and A.3.3.2.

5.1.3.1.2 Flow rate. The selection of the valve, consistent with container size, construction and pressure limitations, shall be based on a minimum flow rate equal to 0.04 times the internal volume of the container per minute. The valve selection shall also account for altitude changes for air transport.

5.1.3.2 Desiccant basket. The container shall be equipped with a desiccant basket mounted to the inner wall of the container and readily accessible from the interior and exterior of the container. The container's desiccant basket shall be designed to contain the quantity of desiccant required as calculated in accordance with MIL-STD-2073-1. All desiccant bags shall be no more than 4 inches from exposure to the closed container's internal air. The desiccant basket shall have a permanent interior captivated cover capable of being opened with common hand tools. When opened, the cover shall not interfere with access to the basket. For exterior desiccant access, the closure shall be waterproof-sealed, captivated and positioned to allow for easy removal of the desiccant bags from the outside of the container. See Table I for the Navy's standard access port. The ability to load and unload a full charge of desiccant from the outside of the container shall be demonstrated during testing.

5.1.3.3 Relative humidity indicator. The container shall incorporate a relative humidity indicator plug conforming to SAE AS26860 with a humidity indicating element relevant to the item. It shall be located as far as possible from the desiccant basket. See Table I for Navy standard humidity indicator.

5.1.3.4 Visual inspection ports. The container shall contain two visual inspection ports that shall conform to Navy drawing 6212706, as specified in Table I, or in accordance with SAE AS26860 humidity indicators with the indicating element removed. The ports shall be located a minimum of 4.0 inches apart. On containers greater than 72 inches high, the ports shall be located at a height less than 60 inches measured from the ground.

5.1.3.5 Records receptacle. An exterior record receptacle shall be provided to permit easy insertion and removal of associated paperwork or records. Records receptacles shall be either a rectangular receptacle (12.0 x 10.0 x 2.0 inches outside dimension) or a cylindrical receptacle (12 x 4 inches diameter – outside diameter), as specified (see 6.2). When the records receptacle penetrates the shell of the container, the pressure boundary integrity of the container shall be maintained. The receptacle shall be located to preclude damage by materials handling equipment and in an attitude such that water shall not collect in it when the cover is open. The closure shall be waterproof-sealed, captivated and positioned to allow for full receptacle access. See Table I for Navy standard records receptacle.

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5.1.3.6 Leak test valve. All sealed containers shall be equipped with an air fill valve in accordance with SAE AS5017, for the purpose of leak testing and fuel vapor checks. The valve shall be located so that the introduced air flows over the desiccant bed and does not impinge on the contained item.

5.2 Base.

5.2.1 Mounting frame. The container design shall prevent shifting of the item within its mounting frame. The provisions for mounting shall be easily accessible to the operator without sacrifice to safety and shall not encumber the proper operation of any tools required to install or remove the item.

5.2.1.1 Restraint conditions. The forces as specified in A.3.11 are to be applied independently. For reference purposes, all vector force directions are relative to an aircraft. If the item is capable of being loaded facing either direction then the forward requirement shall apply to both forward and aft.

5.2.1.2 Mounting frame - VERTREP. When required, the container shall be shipped via VERTREP (see 6.2). Certification shall be obtained through 5.5.1.2 and all requirements of U.S. Army Natick Soldier Research, Development and Engineering Center in Natick, MA. When subject to the tests, any sign of permanent structural deformation in the suspension system or mounting frame shall be cause for rejection.

5.2.2 Suspension system. The container shall have a suspension system consisting of a foam system or a mounting frame with elastomeric or coil shock mounts capable of satisfactory operation over the temperature range of -20 °F through +160 °F. The suspension system shall be resistant to aging and deterioration over the expected life of the container. When loaded, the shock mitigating properties of the suspension system shall meet the dynamic characteristics of 4.3 (see 6.2).

5.2.2.1 Shock isolator markings. Each elastomeric shock mount shall be marked with the manufacturer's name, manufacturing cure date (month and year) permanently molded in the top of the mount material and be plainly visible when the mount is installed. If mount configuration prevents the visual identification specified above, then the cure date (month and year) shall be stamped on the top surface of the flexing element using white waterproof ink conforming to A-A-208. Elastomeric shock mounts shall be no older than one (1) year old when installed in the container. Wire coil mounts shall have the manufacturer's name and part number stamped on the mount assembly and be plainly visible when the mount is installed.

5.2.2.2 Cushioning materials. Cross-linked polyethylene in accordance with A-A-59136 shall be used. Use of any alternate material shall be approved by the design and procuring activities. In addition to these characteristics, the following shall be implemented:

a. When the item is susceptible to damage by electrostatic discharge (ESD), the cushioning material shall have an average decay time of not greater than 2.0 seconds as tested in accordance with MIL-STD-3010, Method 4046. The surface resistance shall be equal to or

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greater than 1.0 by 10^4 ohms, but less than 1.0 by 10^{11} ohms in accordance with ANSI/ESD STM11.11. The resistivity of volume conductive cushioning material shall be 1.0 by 10^4 ohm-centimeter (cm) but less than 1.0 by 10^{11} ohm-cm in accordance with ASTM D991.

b. Cushioning materials shall not emit (outgas) any volatile gas within the greater than 10 percent of the lower explosive limit (LEL).
c. When the size or shape of a particular cushion requires that it be constructed of multiple sections bonded together, the lamination plane(s) shall be perpendicular to the direction of the load on that cushion.
d. Cushioning materials installed with adhesives shall have the capability to be readily removed and replaced without rework to the container.

5.2.3 Fasteners. Fasteners shall be used to provide the required seal using a gasket closure. This seal shall be obtained by easily operated fasteners (closed and opened by hand or by means of a common tool, such as a screwdriver). Deformation of the closure that adversely affects this operation shall be cause for rejection. The appropriate tensile strength over-center latches or equivalent shall be used. They shall have a mechanical provision to eliminate loosening. All fasteners in either open or closed condition shall be protected from damage during shipping and handling. See Table I for the Navy standard latch.

5.2.4 Storage box (if required). The container shall be furnished with an attached storage box to store ancillary components. The ancillary components shall be restrained and, if necessary, shall have shock mitigation while inside the box. The storage box shall not impact design and load test requirements for the container.

5.2.5 Orientation/security seal. The container shall be mechanically keyed for proper orientation and alignment of closure mating surfaces. Mechanical keying is not required if the cover is 180° reversible. The leak test shall be performed with the lid in both positions for 180° reversible covers. The container shall have provisions for the insertion of two leaded wire seals (one each end).

5.2.6 Drain plug. A drain plug (MS20913-6J, SAE AS 4863 J 06 or equivalent) shall be installed in the lowest point of the container shell at the end containing the other service facilities. The plug shall be installed in a bushing and shall be protected from damage from forklift handling. Location of the bushing shall allow for complete drainage when the plug is removed. Installation and removal of the plug shall be from the outside of the container.

5.2.7 Skids. The container shall be constructed with skids that are of the same material as the container shell. The attachment of skids shall not penetrate the interior compartment of the container. They shall be arranged to permit handling by forklift trucks and shall permit easy blocking and bracing in railcar loading and truck loading. Preferred orientation of skids is parallel to the long dimension of the container base, if consistent with meeting forklift capability requirements. Any ramping of skid ends to accommodate skidding of containers shall be kept to a minimum (a 15-degree ramp angle is recommended with a maximum ramp rise of 3⁄4 inch) to minimize blocking and bracing problems. Additionally, the protrusion of skid ends beyond the

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end of the container shall be limited to the minimum length necessary for protecting the container surface and accessories.

5.2.8 Roller interface. The container shall have a solid surface of no less than 5.25 inches around the perimeter of the base that is capable of supporting the loaded container on a roller system during transportation as specified in MIL-STD-3028 and MIL-STD-1791.

5.3 Identification.

5.3.1 Markings. Stenciled markings shall be epoxy base ink conforming to A-A-56032 or A-A-208. Color of marking shall be black conforming to Color #27038 or 27040 of SAE AMS-STD-595. The following markings shall be permanently applied to all containers at the locations and of the sizes indicated.

a. Adjacent to the humidity indicator in 0.5 inch (13 mm) letters:

HUMIDITY INDICATOR

b. Adjacent to the pressure relief valve in 0.5 inch (13 mm) letters:

RELIEF VALVE

c. On all sides, or on at least two locations equally spaced around the circumference of round containers, and above the closure flange in 0.5 inch (13 mm) letters:

CAUTION: RELEASE PRESSURE BEFORE OPENING CONTAINER

d. At the loaded center of balance on both sides of the container, a vertical line 6.0 inches (152 mm) long and 1 inch (25.4 mm) wide with the marking adjacent in 1.0 inch (25 mm) letters:

CENTER OF BALANCE

e. At the loaded center of gravity of the container, a 3.0 inch (76 mm) CG target mark shall appear on the two longitudinal sides of the container with the marking adjacent in 1.0 inch (25.4 mm) letters:



CENTER OF GRAVITY

f. Adjacent to exterior opening of the desiccant basket in 0.25 inch (6 mm) letters:

DESICCANT COMPARTMENT ____ UNITS REQUIRED
(contractor shall insert # of units based on MIL-STD-2073-1 calculations)

g. On all sides of the container, above the horizontal split line in 1.0 inch (25 mm) letters:

DO NOT DROP

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h. On opposite sides of the upper section of the container in 1.0 inch (25 mm) high letters:

REUSABLE CONTAINER - DO NOT DESTROY
RETURN TO SUPPLY WHEN NOT IN USE

i. An arrow 6.0 inches x 1.0 inch shall point to the tiedown rings, eyes or lugs with 2.0 inch (51 mm) letters stating:

TIEDOWN

j. Adjacent to the record receptacle in 1.0 inch (25 mm) letters:

RECORD RECEPTACLE

k. Between each pair of fork lift pockets, arrows pointing to the pockets, and in 1.0 inch (25 mm) high letters stating:

FORK LIFT HERE

l. Adjacent to the leak test valve (when required by 5.4.7) in 0.25 inch (6 mm) letters:

FUEL VAPOR TEST
LEAK TEST

m. An arrow 6.0 inches x 1.0 inch shall point to the lifting device with 1.0 inch (25 mm) letters stating:

LIFT HERE

n. An arrow 6.0 inches x 1.0 inch shall point to the tow rings, eyes, or lugs with 1.0 inch (25 mm) letters stating:

TOW HERE

o. An arrow 6.0 inches x 1.0 inch shall point to the cover lift rings, handles or lugs with 1.0 inch (25 mm) letters stating:

COVER LIFT ONLY

5.3.2 Shipping information. A solid black square 12 inches by 12 inches shall be permanently applied to the upper left corner of two opposite sides of the container using the same epoxy base ink and color as specified in 5.3.1. Additionally, parallel to and above the black square, apply 0.5 inch (13 mm) letters stating:

SHIPPING INFORMATION ONLY

5.3.3 Nameplate. A nameplate conforming to MIL-DTL-15024, Type H, (Photosensitive Aluminum, other than foil) shall be mechanically affixed to the service facility end of the container. In addition, nameplate information shall be included as part of the installation instructions (reference 5.4). The nameplate shall include the appropriate UID (Unique Identification) marking in accordance with MIL-STD-130 and Container Nameplate Drawing (80132) 15930. The nameplate shall measure 4.0 x 4.0 inches (102 x 102 mm) and shall include the following information (see Figure 1):

CONTAINER, SHIPPING AND STORAGE

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APPLICATION: (insert name of equipment)

NAVY P/N (80132) XXXXX

NSN (to be provided by the Government)

DESIGN MANUFACTURER P/N (CAGE) XXXXX

CONTRACT NO. (insert proper number)

SERIAL NO. (to be provided by the government)

CONTAINER MANUFACTURER: (insert name of company)

CONTAINER MANUFACTURER P/N (CAGE) XXXXX

UID: (Data matrices per MIL-STD-130)

All information with the exception of contract number shall be photoetched. The contract number shall be photoetched, stamped or engraved.

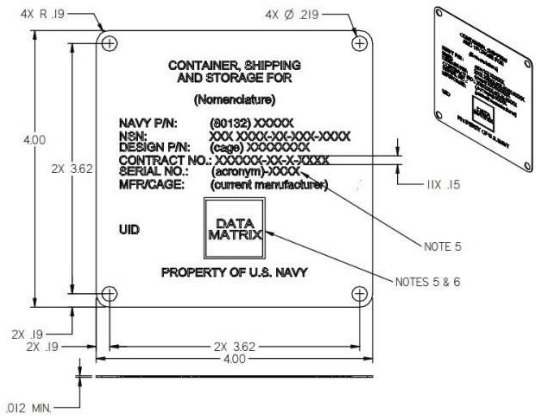


FIGURE 1. Nameplate.

5.3.4 Container serial number. Serial numbers shall be assigned by the government and provided to the manufacturer. The same serial number that appears on the nameplate shall also be engraved/metal stamped into the cover and base extrusions of the container.

5.4 Installation instructions. Laminated installation instructions shall be a permanent part of the container. The instructions shall be bonded to the inner surface of the container, preferably on the bottom half, and positioned for viewing. When the container has a records receptacle, a second set of laminated installation instructions shall be inserted in the receptacle. Adequacy of the installation instructions shall be demonstrated during the fit and compatibility test of A.3.4.

Installation instructions shall include the following:

a. Container isometric illustration
b. Step-by-step procedure for installation and removal of the item
c. Number of desiccant units required
d. Torque value for container closure fasteners

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e. Interior item mounting hardware (when applicable)
f. Container tare weight
g. Container gross weight
h. Container outer dimensions
i. Container maximum cube
j. All information as shown on nameplate (reference 5.3.3)

5.5 Transportability. The container shall reflect consideration of techniques for truck, rail, air, and ship loading to ensure that applicable DOT requirements are met, and that transportation costs are minimized consistent with safety considerations and container integrity (see MIL-STD-1366 and MIL-STD-3028). Specifically, the container shall be transportable with unrestricted movement via fixed-wing air transport through the Defense Transportation System in accordance with MIL-STD-1366, and via tiltrotor aircraft within aircraft limitations defined by NAVAIR A1-V22AB-CLG-000 and MIL-STD-1366.

5.5.1 Transfer-at-sea. The container shall be designed to be transferred-at-sea, by Underway Replenishment (UNREP) techniques in accordance with NWP 4-01.4 to include Standard Replenishment Alongside Method (STREAM), Vertical Onboard Delivery (VOD), and Vertical Replenishment (VERTREP) for (Carrier, Fixed Wing Aircraft, Nuclear) CVN transportability. The container shall satisfy tests of Table A-I, A.3.10 and A.3.11.

5.5.1.1 CONREP. When a container weighs less than 5700 pounds and largest dimension is less than 80 inches, testing for CONREP shall be applied as specified in Table A-I (see 6.2)

5.5.1.2 VERTREP. When specified, the container shall be certified for External Air Transport (EAT)/Helicopter Sling Loading (HSL) by rotary wing aircraft by the U.S. Army Natick Soldier Research, Development and Engineering Center in Natick, MA (see 6.2). The container shall have lifting provisions with a minimum 2½-inch opening and shall interface with the MK 105 MOD 0 Sling and the MK 109 MOD 1 Sling. See Table I for Navy standard lift rings and shackles. Details for Navy VERTREP are found in NWP 4-01.4. The container cover and base shall be designed to withstand crushing loads associated with slings used during VERTREP operations.

5.6 Provisions. Provisions shall meet the following requirements.

Note: When lifting, tiedown, or towing provisions are interchangeably designed or integrated; all of the shared requirements shall be met. Handle provisions requirements are standalone.

5.6.1 Handle provision characteristics. Handles shall not be used as tiedown provisions.

a. When not in use, non-stationary handles shall either swing/rotate down against or retract into the side or end of the container, regardless of the container's angle relative to horizontal.
b. When in use, non-stationary handles shall stop open at a 90-degree angle ±10 degrees in relation to the surface that possesses the handle when lifting.
c. For a Navy common standard part example for non-stationary handles, see Table I.

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d. Stationary handles shall be designed such that they do not extend beyond the container's outer envelope.
e. Handles shall meet the strength requirements of A.3.10.3.
f. Handles used to specifically lift empty containers; or container covers; or container components shall be clearly labeled to avoid misuse.
g. When multiple handles are required to support manual lifting or movement, they shall be ergonomic and located (in accordance with MIL-STD-1472) around the center of balance of the container; or container cover; or the container component to facilitate lifts.
h. Handles shall not be used for cover weights that approach or exceed 88 pounds (two-person male/female lift). In those instances, covers shall have lifting provisions and be clearly labeled "COVER LIFT ONLY" in the cover design.
i. Handles for cover lift only shall be marked in accordance with 5.3.1.

5.6.2 Lifting provision characteristics. Lifting provisions (also known as hoisting provisions) shall have the following characteristics and be located as follows:

a. Lifting provisions shall pass tests as outline in A.3.10.1, A.3.10.2 and A.3.10.3.
b. Lifting provisions shall be marked in accordance with 5.3.1.
c. Lifting provisions shall meet the minimum lifting strength requirement of each lifting point and be capable of lifting the total gross weight of the container if shipped by single-point suspension as detailed in A.3.10.3.
d. Lifting provisions, used as tiedown provisions, shall meet the tiedown strength requirements of A.3.10.4.
e. Lifting provisions:
    1. Shall be placed as far from the loaded center of balance to provide the greatest handling stability.
    2. Shall not result in sling leg angles with the horizontal of less than 30 degrees (45 degrees preferred).
    3. Shall not be placed more than 120 inches (3048 millimeters) apart.
    4. Shall not have sharp edges on the container that damage the sling legs.
    5. Shall center the loaded container's unit load center of gravity (CG) between the lifting provisions.
    6. Shall be compatible with the MK 109 MOD 1 sling for unbalanced containers.
f. Unless the configuration of the lifting sling is known, strength of the provisions and supporting structure shall be based on sling leg angles of 30 degrees to the horizontal. They shall pass testing requirements of A.3.10.2.
g. Lifting provisions shall be provided both on the base and cover, each capable of carrying the entire weight of the loaded container. This capability shall be demonstrated during handling tests.
h. Lifting provisions shall not protrude beyond the container envelope when not in use.
i. Lifting provisions shall not be placed at locations that require the container closure fittings to carry the lifting loads. If unique logistics considerations require a design where the lifting provisions are located in such a manner that require the container closure fittings to carry the lifting loads, the container closure fittings shall meet the strength requirements of A.3.10.5.

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j. Lifting provisions shall be designed to permit full entry and closure of standard safety hooks without undue distortion of the lines attached to the hooks and shall suspend the container parallel to its normal storage attitude.
k. Interior diameter of the lifting provisions shall be in accordance with MIL-STD-209.
l. All lifting provisions shall be made of solid material. Wire rope or other materials that are capable of fraying, shall not be used for the construction of lifting provisions.
m. Lifting provisions shall be replaceable without damaging or requiring alteration of any other part of the container.
n. For containers requiring at-sea replenishment, lifting provisions shall be centered about the loaded CG and spacing shall be limited to 26 inches to 42 inches apart (container width) and 120 inches apart (adjacent or long side).
o. Pass/fail criteria shall be in accordance with 5.6.5.

5.6.3 Tiedown provisions. The container shall be constructed with tiedowns for transport by land, sea and air. The tiedowns shall be configured to limit side loads to any handling equipment, including the hook. When tie downs are used as lifting fittings, the tiedowns shall be also be tested for lifting fitting conformance as specified in A.3.10.2. Tiedowns shall pass strength requirements of A.3.10.4. All tiedown provisions are considered lift points. Pass/fail criteria shall be in accordance with 5.6.5. The interior diameter of the lifting provisions shall be in accordance with MIL-STD-209.

5.6.4 Towing provisions. The container shall have towing capability and shall meet the testing requirements specified in A.3.12. Towing provisions shall be vertically and horizontally centered and be *at or below* the horizontal CG, in both the loaded and unloaded configurations. Towing provisions shall be marked as specified in 5.3.1. Pass/fail criteria shall be in accordance with 5.6.5. The interior diameter of the towing provisions shall be in accordance with MIL-STD-209.

5.6.5 Pass/fail criteria. Any observation of the following (for any lifting, tiedown or towing provision) shall be cause for rejection:

a. Failure of any part of the lifting, closure, or tiedown structure.
b. Permanent deformation (for example, necking) of any part of the structure, supporting or attachment point, that renders the container unsafe or unsuitable for continued handling.
c. Inadequate geometry of tiedown or lifting fitting attachment point, or inability of the provision to properly interface with identified handling equipment in a safe manner.
d. Inability of handling equipment hooks to interface with the container provisions.
e. Inability of the provision to swivel or maneuver in a manner where there is no side loading imparted to the handling equipment hooks.
f. Inability for the provisions to maintain acceptable sling leg angles when lifted with Navy handling equipment.
g. Inability for the container to remain in stable horizontal attitude.

5.7 Shore based. The container shall be designed to be compatible with designated and approved handling equipment.

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5.7.1 Forklift capability. At a minimum, the container shall have side forklift pockets. No additional blocking shall be necessary. The forklift shall not permanently deform the container during any lifting and movement operations.

a. Forklift pockets shall measure 15.0 inches wide x 4.0 inches high, as a minimum.
b. Container forklift pocket openings shall be spaced 15 - 40 inches apart (on center).
c. Forklift tines shall engage ⅔ the length of the container through each set of forklift pockets.
d. The loaded and unloaded container's CG shall be centered between forklift pockets.
e. The forklift pocket shall be an encapsulating tube to the maximum extent practical, while still accommodating pallet jacks in accordance with 5.7.3.
f. Performance testing shall be in accordance with A.3.12 (forklift truck compatibility test).

5.7.2 MK 45 handlift truck interface. The ends of the container shall be designed to interface with the MK 45 handtruck, reference drawing (10001) 2642780. These features shall be tested as detailed in A.3.13.

5.7.3 Pallet jack interface. The container shall be capable of Pallet Jack interface on all sides. The opening height shall be a minimum of 4.0 inches. If the container is over 1,000 pounds, this requirement is not required. Pallet jack compatibility shall be tested as detailed in A.3.14. Any inability to insert or remove the pallet jack shall or instability during testing be cause for rejection.

5.8 Mass simulator design. Whenever possible, the actual item shall be used for the container development and verification phases. In the event the actual item is not available, a mass simulator representative of the actual item shall be designed and manufactured for same. The mass simulator shall include the following criteria: (Note: A steel pipe is not a viable test shape.)

a. Envelope dimensions including external features, such as pins, antennas, and lanyards, that have a bearing on the package design
b. Hard zones for forward, CG and aft locations for mounting of test instrumentation. Mounting points or external hard zone area geometry and strength (Bending or shear moments along the actual item's entire length or skin thicknesses representing soft zones are not acceptable mounting points.)
c. Surface finishes that require special handling provisions
d. Weight, CG, and radii of gyration in the three principal axes for the overall mass simulator as well as for each section of the Mass Simulator
e. Internal joints and shock mounts identical to the actual item
f. Identified stations where fragility readings are critical and allowable levels of fragility at those locations

When specified, the developed mass simulator design shall be documented on the Contract Data Requirements List (CDRL) and a Product Level Technical Data Package (TDP) shall be delivered (see 6.3).

5.9 Finish. All surfaces other than threaded or working surfaces shall be processed as indicated in 5.9.

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5.9.1 Aluminum. Aluminum shall be cleaned in accordance with MIL-DTL-5541. Further chemical finish or painting is not required.

5.9.2 Surface finishing. Aluminum containers shall be glass bead blasted on exterior shell surfaces to provide a uniform matte finish that shall hide minor surface imperfections. Size #8 glass beads in accordance with MIL-PRF-9954 shall be used with direct air pressure ranging from 50 to 70 psig. A separator shall be used to provide a reclaim efficiency of 90 percent (a minimum of 90 percent by weight of the original material shall be in the reclaim). The glass bead media shall be maintained in a condition free from contamination of other metals.

5.9.3 Plating. Finish shall be electrodeposited alkaline zinc-nickel alloy in accordance with ASTM B841 Class 1, Type B, Grade 3 or zinc plating in accordance with ASTM B633, or hot dip galvanize, in accordance with ASTM A153/A153M.

5.9.4 Stainless steel. Stainless steel shall be passivated in accordance with SAE AMS2700, ASTM A380/A380M and ASTM A967/A967M.

5.10 Workmanship. The container shall be manufactured free of any defects that affect durability, strength, safety and serviceability. Any distortion of the container shell surface due to manufacturing processes that exceeds ±1.0 percent of container length or ±½ inch (whichever is less) beyond the designed normal plane shall be cause for rejection. In addition, any manufacturing-caused distortion that affects serviceability and/or function of the container shall be cause for rejection regardless of its magnitude. All sharp edges shall be broken to prevent injury to personnel. All bolted support structures shall have flat mating surfaces that are free of burrs or weld spatter.

5.10.1 Welding. Welding and welder qualifications shall be in accordance with AWS D1.1/D1.1M for steel, AWS D1.2/D1.2M or MIL-W-22248, Class 4, for aluminum, AWS D1.3/D1.3M for sheet steel and AWS D1.6/D1.6M for stainless steel. Proper controls shall be used to prevent melt through or burn through. For aluminum, filler for welding 6061 alloy shall be 4043; filler for welding 5000 series alloys to themselves or to 6061 shall be 5356 or 5556.

5.10.1.1 Visual inspection criteria for aluminum welds. Weld visual inspection shall include 5X (5 power) magnification when a suspect condition is to be examined beyond the capability of normal vision. Weld acceptable criteria include:

1. Lack of surface cracks
2. Complete fusion between adjacent weld metal layers
3. Complete fusion between the weld filler and base metal
4. Weld profiles conforming to AWS D1.2/D1.2M
5. Undercuts not exceeding the lesser of 10% plate thickness or 1/16″
6. Overlap not exceeding 2T in any 6T length, where T = thickness
7. Continuous fillet welds, undersized by 1/16″ nominal, not exceeding 10% overall weld length
8. Shrinkage contained entirely within the crater area of intermittent fillet welds
9. Lack of porosity

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When specified, critical and major welds shall be subjected to additional non-destructive inspection (NDI) using dye penetrant to confirm lack of surface cracks (see 6.2).

5.10.1.2 Visual inspection criteria for steel welds. The pass/fail criteria for ASTM A514/A514M, A517/A517M steel weld shall be based on visual inspection performed no more than 48 hours after completion of the weld, but no sooner than immediately after completed welds have cooled to ambient temperature. All steel welds shall satisfy the requirements of AWS D1.1/D1.1M, Table 6.1.

Convex, flat, or slightly concave fillet welds are acceptable as specified in AWS D1.1/D1.1M. Unacceptable fillet weld profiles are defined AWS D1.1/D1.1M. Groove or butt welds shall be made with minimum face reinforcement unless otherwise specified. Face reinforcement shall not exceed 1/8″ in height for butt and corner joints. All welds shall have a gradual transition to the plane of the base-metal surfaces with transition areas free from undercut except as permitted by this code.

Weld visual inspection shall include 5X (5 power) magnification when a suspect condition is to be examined beyond the capability of normal vision. A weld shall be acceptable by visual inspection if it exhibits:

1. An absence of cracks, regardless of size or location.
2. Complete fusion between adjacent layers of weld metal and between weld filler and base metal.
3. Filled craters providing specified weld size, with the exception of intermittent fillet weld end points outside of their effective length.
4. Undersized welds not exceeding 10 percent of the weld length. On web-to-flange welds, end underrun shall be prohibited for a length equal to twice the width of the flange
5. Undercut not exceeding 1/32″ except for primary members, where undercut shall not exceed 0.01″ in depth when the weld is transverse to the direction of tensile stress under any design loading conditions.
6. No visible piping porosity in groove welds and butt joints transverse to the direction of computed tensile stress. For all other groove and fillet welds, the frequency of piping porosity shall not exceed one instance in 4″ of weld length and the maximum diameter shall not exceed 3/32″.

For welds subject to additional NDI (see 6.2), including magnetic particle and liquid penetrant, all NDI methods including equipment requirements and qualifications, personnel qualifications, and operating methods shall be in accordance with AWS D1.1/D1.1M, Section 6, Inspection.

a. Magnetic particle testing shall be in accordance with ASTM E1444/E1444M with standards of acceptance defined by AWS D1.1/D1.1M.
b. Dye penetrant tests to detect surface discontinuities shall be in accordance with ASTM E165/E165M with standards of acceptance defined by AWS D1.1/D1.1M.

5.10.1.3 Cleaning. Parts to be welded shall be cleaned to remove surface soils, except that for aluminum, uninhibited alkaline solutions such as sodium hydroxide shall not be used.

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5.10.1.4 Deoxidizing. Aluminum parts shall be chemically deoxidized no more than 10 days prior to welding to remove thick surface oxide films. (If the material or work is exposed to an outdoor environment, this time limit shall be reduced to a maximum of 3 days, to account for atmospheric effects.) Mechanical cleaning methods shall be applied just prior to the actual start of welding, to remove any reoxidation or residual thin oxide films.

5.10.1.5 Chemical treatment. An acid deoxidizing treatment shall be applied by either immersion or brushing/wiping. Deoxidizers acceptable for use shall include nitric acid, sulfuric chromic, phosphoric chromic or equivalent acid solutions. The solution shall cover 100 percent of the weld area and extend at least 2 inches from the weld site (or end at any closer adjacent edge when space is unavailable).

5.10.1.6 Mechanical means. Immediately prior to welding, mechanical cleaning methods shall be applied to previously chemically-treated weld areas to remove all residual or reformed oxides. Acceptable methods of mechanical cleaning include stainless steel wire brushing, scraping, filing, or sanding. However, abrasives containing iron and its oxides, steel wool and wire, and copper alloy based wire shall not be used. Mechanical methods shall remove all residual oxide films without weld surface degradation.

5.11 Verification. Verification of the requirements detailed in this document or contract shall be via direct test, measurement or engineering analyses as indicated in Table II. Any failure that occurs during government testing that impedes the ability to continue to test, the container shall be shipped back to the manufacturer for rework, repair or redesign (as applicable) and all affected tests shall be repeated.

TABLE II. Verification requirements matrix.

Paragraph	Requirement Characteristics	Verification By				
		I	S	A	T	TG
4.1	General Requirements	◊				
4.1.3	Pressure and Vacuum Retention	◊				◊
4.1.4	Drainage	◊				
4.1.6	Container Structure Material	◊				
4.2	Fit and Compatibility					◊
4.3	Dynamic Characteristics	◊				
5.1	**COVER**					

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TABLE II. Verification requirements matrix – Continued.

Paragraph	Requirement Characteristics	Verification By				
		I	S	A	T	TG
5.1.1	Stacking Feature	◊				
5.1.2	Sealing					
5.1.3	Service and Maintenance Facilities	◊				
5.1.3.1	Breather Valve	◊				
5.1.3.1.1	Cracking & Sealing Pressure	◊				
5.1.3.1.2	Flow Rate			◊		
5.1.3.2	Desiccant Basket	◊				
5.1.3.3	Relative Humidity Indicator	◊				
5.1.3.4	Visual Inspection Ports	◊				
5.1.3.5	Record Receptacle	◊				
5.1.3.6	Leak Test Valve	◊				
5.2	**BASE**					
5.2.1	Mounting Frame	◊				◊
5.2.1.1	Restraint Conditions					◊
5.2.1.2	Mounting Frame - VERTREP					◊
5.2.2	Suspension System	◊				
5.2.2.1	Shock Isolator Markings	◊				
5.2.2.2	Cushioning Materials	◊				
5.2.3	Fasteners	◊				
5.2.4	Storage Box	◊				
5.2.5	Orientation/Security Seal	◊				

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TABLE II. Verification requirements matrix – Continued.

Paragraph	Requirement Characteristics	Verification By				
		I	S	A	T	TG
5.2.6	Drain Plug	◊				
5.2.7	Skids	◊				
5.2.8	Roller Interface	◊				
5.3.1	Markings	◊				
5.3.2	Shipping Information	◊				
5.3.3	Nameplate	◊				
5.3.4	Container Serial Number	◊				
5.4	Installation Instructions	◊				
5.5	**TRANSPORTABILITY**	◊				
5.5.1	Transfer-at-Sea	◊				
5.5.1.1	CONREP					◊
5.5.1.2	VERTREP					◊
5.6	**PROVISIONS**					
5.6.1	Handle Provision Characteristics	◊				
5.6.2	Lifting Provision Characteristics	◊				
5.6.3	Tiedown Provisions	◊				
5.6.4	Towing Provisions	◊				
5.7	**SHORE BASED**					
5.7.1	Forklift Capability	◊				
5.7.2	MK 45 Handlift Truck Interface	◊				
5.7.3	Pallet Jack Interface	◊				

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TABLE II. Verification requirements matrix – Continued.

Paragraph	Requirement Characteristics	Verification By				
		I	S	A	T	TG
5.8	Mass Simulator Design	◊				
5.9	**FINISH**					
5.9.2	Surface Finishing	◊				
5.10	**WORKMANSHIP**	◊				
5.10.1.1	Visual Inspection Criteria for Aluminum Welds	◊				
5.10.1.2	Visual Inspection Criteria For Steel Welds	◊				
A.3.1.1	Instrumentation					◊
A.3.2	Inspection					◊
A.3.3.1	Leak test				◊	◊
A.3.3.2	Vacuum Test				◊	◊
A.3.4	Fit and Compatibility Test					◊
A.3.5	**VIBRATION**					
A.3.5.2	Sine Vibration			◊		◊
A.3.5.1.1	Resonance Survey			◊		◊
A.3.5.1.2	Sine Dwell					◊
A.3.5.2	Random Vibration					◊
A.3.5.2.1	Truck Transportation Vibration			◊		◊
A.3.5.2.2	Underslung Cargo Vibration			◊		◊
A.3.6	**MECHANICAL SHOCK**					
A.3.6.1	Cornerwise-Drop (Rotational) Test			◊		◊

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TABLE II. Verification requirements matrix – Continued.

Paragraph	Requirement Characteristics	Verification By				
		I	S	A	T	TG
A.3.6.2	Edgewise-Drop (Rotational) Test			◊		◊
A.3.6.3	Impact Test			◊		◊
A.3.6.5	Flat Bottom Drop Test			◊		◊
A.3.6.6	Repetitive Shock Test					◊
A.3.7	Tipover Test					◊
A.3.8	Load Stability			◊		◊
A.3.9	Superimposed load			◊		◊
A.3.10	Lifting Fittings, Closure Fittings, and Tiedown Attachment Points	◊				
A.3.10.1	Lifting Fittings Interface Test					◊
A.3.10.2	Lifting Fittings Strength Test					◊
A.3.10.3	Single Lifting Fitting Strength Test					◊
A.3.10.4	Tiedown Strength Test					◊
A.3.10.5	Closure Fitting Strength Test					◊
A.3.11	Mount and Structure Load Test			◊		◊
A.3.12	Forklift Truck Compatibility Test					◊
A.3.13	Handlift Truck MK 45 Compatibility					◊
A.3.13.1	Static Overload					◊
A.3.13.2	Shock Test					◊
A.3.13.3	Rolling Test					◊
A.3.14	Pallet Jack Compatibility Test					◊

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I – Inspection
S – Similarity
A – Analysis (Engineering)
T – Testing (Manufacturer)
TG – Testing (Government)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended for NAVAIR Program Offices and NAVSUP to provide tailored requirements for design and testing of metal specialized reusable shipping containers for aviation repairables.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.
b. Item sensitivities and special shipping requirements (see 4.1)
c. Reusable container product level technical data package (see 4.1)
d. Loading and unloading times, if required (see 5.1)
e. Records receptacle type (rectangular or cylindrical) (see 5.1.4.5)
f. Non-operational fragility limit of the item (see 5.2.2, A.3.5 and A.3.6)
g. Selection of random or sinusoidal vibration (see 5.5.5 and A.3.5)
h. VERTREP Certification, if required (see 5.2.1.1, 5.2.1.2 and 5.5.1.2)
i. Mass simulator design and delivery, if required (see 5.8)
j. Temperature extreme testing, if required (see 4.1.2, 5.2.2 and A.3.6.4)
k. Modes of transportation that item will be exposed (see 5.5 and 5.7)
l. Non-Destructive Inspection (test and documentation) requirements (see 5.10.1.1 and 5.10.1.2)

6.3 Associated Data Item Descriptions (DIDs). This standard has been assigned an Acquisition Management Systems Control (AMSC) number authorizing it as the source document for the following DIDs. When it is necessary to obtain the data, the applicable DIDs must be listed on the Contract Data Requirements List (DD Form 1423).

DID Number	DID Title
DI-SESS-80640	Request for Variance (RFV)
DI-PACK-82384	Specialized Reusable Container Design Candidate Technical Data Package (TDP)

The above DIDs were current as of the date of this standard. The ASSIST database should be researched at https://quicksearch.dla.mil to ensure that only current and approved DIDs are cited on the DD Form 1423.

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6.4 International standardization agreement implementation. This standard implements STANAG 2828 Military Pallets, Packages and Containers and STANAG 2829 Materials Handling Equipment. When changes to, revision, or cancellation of this standard are proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://quicksearch.dla.mil.

6.5 Subject term (key word) listing.

Shipping
Storage
Testing

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TEST METHODS

A.1 SCOPE

A.1.1 Scope. This Appendix details the test methods to be performed on specialized reusable containers used for the protection of naval aviation repairables. This Appendix is a mandatory part of the standard. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in the Appendix of this standard. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited Appendix A of this standard, whether or not they are listed.

A.2.2 Standards. The following standards form a part of this Appendix to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-209	-	Lifting and Tiedown Provisions
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-913	-	Requirements for the Certification of Sling Loaded Military Equipment for External Transportation by Department of Defense Helicopter
MIL-STD-1791	-	Designing for Internal Aerial Delivery in Fixed Wing Aircraft
MIL-STD-3010	-	Test Procedures for Packaging Materials and Containers

(Copies of these documents are available at https://quicksearch.dla.mil.)

A.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D999	-	Standard Test Methods for Vibration Testing of Shipping Containers

(Copies of this document are available online at http://www.astm.org.)

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HANDBOOK

DEF STAN 00-35 Part 3 - Environmental Handbook for Defense Materiel Environmental Test Methods

(Copies of this document are available from https://standards.globalspec.com.)

A.3 VERIFICATION AND TEST REQUIREMENTS

A.3.1 Mass simulator. The actual item or developed mass simulator (see 5.8) shall be used in all verification tests.

A.3.1.1 Instrumentation. The item or mass simulator shall be instrumented, at a minimum, at the forward, CG, and aft locations. Data shall be collected in all three principle directions (vertical, transverse and longitudinal) during testing to assure that established fragility levels and vibration allowances are not exceeded. Shock time histories (Gs vs. time) in the major directions shall be recorded on the Mass Simulator at the forward, CG, aft locations and at zones where the fragility is clearly defined. The magnitude of the acceleration shall be determined by the vector addition of the three (3) components of the triaxial accelerometer whose sensitive axes are mounted perpendicular to one another. The data shall be analyzed point by point along the common time base to determine the maximum resultant peak acceleration. The maximum resultant peak acceleration shall not exceed the approved shipping/non-operational fragility limit of the item. For uniquely configured or irregularly shaped items, and in cases where known fragility points have not been established, engineering analysis shall be used in selecting instrumentation locations.

A.3.2 Inspection. The container shall be inspected to the appropriate drawing package as well as the features as described in 5.1 through 5.4. Any discrepancies shall be noted and shall be cause for rejection.

A.3.3 Leakage integrity. The test for leakage integrity shall be accomplished before the first test, following each series of shock, vibration, and handling tests and upon completion of all testing. Pass/fail criteria for testing shall be as specified in A.3.3.1.

A.3.3.1 Leak test. The container shall be prepared for testing by sealing all breathing devices and inserting pressure fittings and gages. The container shall be closed and sealed, as it would be for shipping. The test shall be conducted by increasing and decreasing the internal pressure so that the desired test pressure is obtained. The test pressure shall be 1.00 +0.05/-0.00 psig. Any container which has a change in pressure in excess of 0.05 psig in 30 minutes, corrected for variations in air temperature and ambient pressure, shall be cause for rejection. Additional tests shall be performed to pinpoint the leakage. In addition, any temporary deformation of the container shell which is more than three-quarters of an inch beyond its position at ambient pressure shall also be cause for rejection. Permanent deformation shall be defined as any deformation which is not self-correcting upon relief of the pressure differential and that deformation that restricts the disassembly, reassembly and subsequent reuse of the container (form, fit, function).

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WARNING:

- There is potential for the container to explode or fasteners to fail during the Leak/Vacuum Test!
- Use protective barriers to avoid injury to personnel.

A.3.3.2 Vacuum test. The container shall be prepared for testing by sealing all breathing devices and inserting suitable pressurizing fittings and gages. The container shall be closed and sealed, as it would be for shipping. The test shall be conducted by increasing and decreasing the internal pressure so that the desired test pressure is obtained. The test pressure shall be -1.00 +0.00/-0.05 psig. Any container which has a change in pressure in excess of 0.05 psig in 30 minutes, corrected for variations in air temperature and ambient pressure, shall be cause for rejection. Additional tests shall be performed to pinpoint the leakage. In addition, any temporary deformation of the container shell which is more than three-quarters of an inch beyond its position at ambient pressure shall also be cause for rejection. Permanent deformation shall be defined as any deformation which is not self-correcting upon relief of the pressure differential and that deformation that restricts the disassembly, reassembly and subsequent reuse of the container (form, fit, function). The vacuum test shall be the last test performed.

A.3.4 Fit and compatibility test. This test shall be conducted by properly positioning and securing the actual item in the container. The cover shall be installed on the base, all restraining devices fastened, desiccant loaded, and otherwise the container completely secured for shipment. If the container possesses installation instructions, they shall take precedence over any other loading sequence, and shall be followed to verify fit when performing both the loading and unloading sequence. The container shall then be unloaded. Only the material handling equipment available at points of loading and unloading shall be used. Details of the number of personnel required, man hours expended and all handling equipment used shall be recorded. The container and the item or mass simulator shall be representative of the final configuration. In addition, the ability to load and unload a full charge of desiccant from the outside of the container shall be demonstrated. Pass/fail criteria for testing shall be as specified in 4.2.

A.3.5 Vibration. Containers that employ special energy-absorbing devices for shock isolation purposes, or that have sufficient structural flexibility with resonances below 50 hertz, shall be designed to satisfactorily survive exposure to the appropriate vibration test (see 5.2.5 and 6.2). Mass simulators as defined in A.3.1 shall be used. The container and associated isolation system shall meet the requirements of 4.3 and the following:

a. For random vibration testing, clearly defined vibration fragility levels shall be provided by the developer of the item being tested. Fragility levels shall represent the intended mode of transportation and corresponding test input levels. Peak fragility levels shall be defined in g^2/hertz (Hz) breakpoints at defined frequencies. MIL-STD-810, Method 514.6 shall be used in known input profiles for environmental vibration exposures.

b. The isolation system's natural resonant frequency in the major translational modes of vibration shall be above 7.5 Hz.

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c. For containers employing elastomeric mounts, vibration testing shall be interrupted when the external temperature of any elastomeric mount exceeds the manufacturer's recommended maximum temperature. When manufacturer temperatures are undefined, testing shall be interrupted when mount temperature exceeds 120 °F and resume when the mount temperature has cooled to 90 °F or lower.

The same container shall be used for all vibration and mechanical shock tests. All vibration tests shall be performed prior to mechanical shock tests of A.3.6.

A.3.5.1 Sine vibration.

A.3.5.1.1 Resonance survey. The loaded container shall be placed in its normal shipping position, and rigidly attached to the vibration exciter. The item or mass simulator shall be instrumented as required in A.3.1.1. A search for resonance shall be conducted by applying a sinusoidal input profile from 5 to 50 Hz; with a constant 0.125-inch double amplitude input displacement from 5 to 12.5 Hz, switching to a constant 1 g input from 12.5 to 50 Hz. The sweep rate shall be ½ octave per minute. Transmissibility data shall be obtained over the frequency test range. This test shall be performed before and after the dwell test of A.3.5.1.2.

A.3.5.1.2 Sine dwell. A dwell test of 30 minutes total duration shall be conducted at the predominant sine resonance. The input excitation for the dwell test shall be equal to that used during the sweep test at the indicated frequency

Note:

1. The sweep and dwell tests the test may be interrupted to prevent excessive temperature rise of resilient materials.
2. An adjustment in frequency may be necessary during the course of the test to compensate for shifts in resonant frequency due to temperature rise of the isolator(s).
3. Dwell tests shall not be performed on plastic foam or cushion isolation systems.

A.3.5.2 Random vibration.

A.3.5.2.1 Truck transportation. Testing shall be in accordance with MIL-STD-810 (Method 514.6, Annex C, Category 4, Truck Transportation Over U.S. Highways). Testing shall be in the vertical direction only, in its normal shipping position. The test frequency range shall be 5 to 500 Hz.

A.3.5.2.2 Underslung cargo. Testing shall be in accordance with DEF STAN 00-35 Part 3 Chapter 2-01, Figure A-18 Vibration Test Severity for Helicopter Underslung Loads. Testing shall be in the vertical direction only, in its normal shipping position.

A.3.6 Mechanical shock. Shock tests shall be at the Level A severity. The container and associated isolation system shall meet the requirements of 4.3.

A.3.6.1 Cornerwise-drop (rotational) test. This test shall be conducted in accordance with MIL-STD-3010, Method 5005. Level A protection shall be used.

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A.3.6.2 Edgewise-drop (rotational) test. This test shall be conducted in accordance MIL-STD-3010, Method 5008. Level A protection shall be used.

A.3.6.3 Impact test. The test shall be conducted in accordance MIL-STD-3010, Method 5023. Level A protection shall be used. All four sides of the container shall be impacted.

A.3.6.4 Transfer-at-sea shock test. Containers planned for transfer-at-sea shall be designed to withstand the shock test associated with the appropriate mode of transfer, as shown in Table A-I, at expected temperature extremes.

TABLE A-I. Shock test for transfers.

Test	CONREP with STREAM	VERTREP 1/	Dock side only
Impact velocity 2/	7.0 feet/second	7.0 feet/second	5.0 feet/second
Flat bottom drop	18.0"	18.0"	18.0"

NOTES:
1/ Impact velocity does not account for any sea state conditions.
2/ Impact velocity testing shall be performed on ends and sides.

A.3.6.5 Flat bottom drop. The container in its normal shipping condition shall be subjected to a free-fall drop test on its base by raising it vertically from its normal storage position and then allowing it to fall freely onto a rigid surface from a height of 18 inches at ambient temperature.

A.3.6.6 Repetitive shock test. This test shall be conducted in accordance with MIL-STD-3010, Method 5019 or ASTM D999, Method A2. The table frequency shall be adjusted to ensure that one edge of the container leaves the table not less than 1⁄16 inch on each cycle. This test shall be conducted at ambient temperature. For containers possessing elastomeric shock mounts, A.3.5c shall apply.

A.3.7 Tipover test. One edge of the loaded container shall be raised until the base forms a 20° (± 0.5°) angle with the floor. The load shall be held at a 20° angle for 2 minutes. Pass/fail criteria for testing shall be as specified in 5.1.1. This configuration shall be tested in two mutually perpendicular planes.

A.3.8 Load stability. The base shall be placed on the cover and tilted 20° (± 0.5°) from the horizontal for a period of 2 minutes. The test shall be conducted once with the base empty and once with the base loaded. Each configuration shall be tested in two mutually perpendicular planes. Pass/fail criteria for testing shall be as specified in 5.1.1. Additionally, the inability of the container's stacking features to create a stable arrangement constitutes cause for rejection.

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A.3.9 Superimposed load. Proof of adequate stacking strength shall be tested in accordance with MIL-STD-3010, Method 5016. Pass/fail criteria for testing shall be as specified in 5.1.1.

A.3.10 Lifting fittings, closure fittings, and tiedown attachment points. When the lifting fittings are also used as tiedowns, the lifting fittings shall be additionally tested for tiedown conformance as specified in A.3.10.4. Pass/fail criteria for testing shall be as specified in 5.6.5.

A.3.10.1 Lifting fittings interface test. In the loaded and unloaded conditions, the container shall be lifted with an appropriate rated sling. The container shall maintain a level attitude (±5°) from horizontal in the lifted condition.

A.3.10.2 Lifting fittings strength test. The lifting fittings on containers shall be tested to load limits as specified in MIL-STD-209 or MIL-STD-913 for external air transport (EAT)/helicopter sling load (HSL), or five times the gross weight of loaded container (whichever is greatest). An appropriately rated sling set shall be attached to the lifting fittings, the load lifted off the ground, and held for a period of five minutes. When the intended sling or beam is unknown, then the provisions shall be tested at 30 degrees from the horizontal. For an alternative test method, the container shall be restrained and the lifting loads shall be applied to the container to simulate the conditions above. Following the test, the container lifting fittings shall be inspected.

A.3.10.3 Single lifting fitting strength test. Each lifting provision on the container shall be individually tested by lifting the loaded container and keeping it suspended entirely off any surface for a period of 5 minutes. This shall include “cover lift only” provisions.

A.3.10.4 Tiedown strength test. A load shall be applied to each tiedown provision in the same way that it would be applied in service. In the absence of clearly identified tiedown procedures, the load shall be applied at an angle of 45° (±3°) downward from the horizontal and simultaneously 45° (±3°) outboard from the container surface. The load shall be applied in each of the directions that result from shipment. The duration of the test shall be 5 minutes in each of the test directions. The combined tiedown features shall restrain the container in the shipping orientation without loss of structural integrity when subjected to the following loads applied statically and independently: 3.0 g forward/aft, 1.5 g laterally, and 2.0 g downward. For the four tiedowns on containers or unit loads under 108 inches long, each tiedown feature shall be separately loaded to 1.5 times the container weight forward, aft, and perpendicular to the container. Alternatively, a single test with the appropriate vectored components is acceptable as a substitute for the independent direction tests.

A.3.10.5 Closure fitting strength test. The container shall be prepared for the test by engaging all corner closure fittings and disengaging every other non-corner closure fitting. The engaged closure fittings shall then be overloaded to five times the required lifting load of the container closures. This shall be accomplished by overloading the base of the container to five times the lifting load, attaching a sling, lifting beam, or other device representing normal procedure to the lifting provisions, and lifting the container until it is free of the support. The container shall be suspended entirely off any surface for 5 minutes and then examined for

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damage. Alternatively, the container base shall be suitably restrained and the lifting loads applied to the container in a manner that simulates the conditions above.

A.3.11 Mount and structure load test. This test shall be conducted in accordance with MIL-STD-1791. Each load shall be applied independently to the suspended structure. The load shall be held for 3 minutes. If the item is capable of being loaded facing either direction, then the forward requirement applies to both forward and aft. Pass/fail criteria for testing shall be as specified in 5.2.1. Additionally, the suspension system or inner frame shall not show any permanent structural deformation. Load restraints shall be as follows:

Up	3.0g
Down	4.5g
Forward	3.0g
Lateral	1.5g

A.3.12 Forklift truck compatibility test. This test shall be conducted in accordance with MIL-STD-3010, Method 5011, Lifting and Transporting by Forklift Truck Test, Pushing Test and Towing Test. When the container exceeds 45 inches in width or 91 inches in length, the pushing and towing tests shall be repeated with one end of the container lifted off the ground about 6 inches by the tips of the forks inserted between the skids. Pass/fail criteria for testing shall be as specified in 5.7.1.

A.3.13 Handlift truck MK 45 compatibility.

A.3.13.1 Static overload. The container shall be weighted to three times its maximum normal load. Each end of the container shall be supported by a MK 45 handlift truck or test fixture that simulates the loading conditions, and raised to a height of 3 inches (-0, +0.5) above ground level for 5 minutes. Pass/fail criteria for testing shall be as specified in 5.7.2. For containers stowed aboard ship, the test shall be run two times; once each in the level position and once at a 15° (±1°) angle with one end elevated.

A.3.13.2 Shock test. The container shall be weighted to three times its maximum normal load. Each end of the container shall be supported by a MK 45 handlift truck or test fixture that simulates the loading conditions, and raised to a height of 3 inches (-0, +0.5) above ground level. The entire assembly shall then be moved longitudinally at a walking speed of not less than 2 feet per second up a 30° (±2°) ramp to a height of 1 inch and then allowed to fall abruptly onto a rigid surface. The entire container shall be tested six times in this manner; three times with the forward end first, then three times with the aft end first. Pass/fail criteria for testing shall be as specified in 5.7.2.

A.3.13.3 Rolling test. The container shall be weighted to its maximum normal load. Each end of the container shall be supported by a MK 45 handlift truck, or test fixture that simulates the loading conditions, and raised to a height of 3 inches (-0, +0.5) above ground level. It shall then be moved longitudinally at a speed of not less than 4 feet per second on a dry, smooth surface with a coefficient of sliding friction of 0.6. After 20 feet of movement, a sharp 45° (±5°) turn to the left shall be made while the trailing brakes are engaged. This test shall be repeated

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with a sharp 45° (±5°) turn to the right while the brakes of the trailing handlift truck are engaged. Pass/fail criteria for testing shall be as specified in 5.7.2.

A.3.14 Pallet jack compatibility test. This test shall be conducted in accordance with MIL-STD-3010, Method 5011, Pallet Truck Test. All sides with fork pockets shall be tested. Pass/fail criteria for testing shall be as specified in 5.7.3.

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CONCLUDING MATERIAL

Custodian:
Navy-AS

Preparing activity:
Navy-AS
Project PACK-2022-006

Review activity:
Navy-SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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